



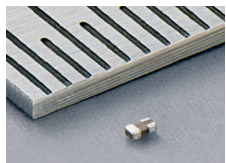
change allows engineers to streamline their designs and reduce costs and effort. This qualification update means that engineers who previously included additional isolation in their designs can now eliminate these components. It also offers the opportunity for those designs that only met basic isolation requirements to be re-qualified as reinforced isolated systems without altering the BoM.

Melexis

<https://www.melexis.com>

Antenna interference countermeasures for wearables

Murata Manufacturing Co., Ltd has launched Radisol, a product designed to enhance antenna isolation and radiation efficiency by reducing interference between closely positioned antennae



with minimal insertion loss. This product is aimed at the design needs of devices like

smartphones and wearables, offering advantages such as reduced power consumption, smaller size, and improved communication quality. It minimises antennae interference without significantly affecting the passband, thus boosting radiation efficiency and reducing power consumption. It serves as a specific filter circuit for each communication band, integrating a capacitor and two inductors within one chip to provide band-stop filter characteristics without noticeable insertion loss. Radisol delivers effective antenna isolation with low insertion loss and improved system efficiency and integration.

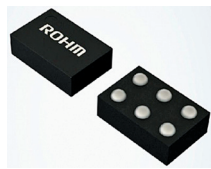
Murata Manufacturing Co., Ltd

<https://www.murata.com>

Industry's smallest CMOS op amp for smartphones

ROHM has introduced the TLR377GYZ, an ultra-compact CMOS operational amplifier (op amp) designed for 1.8V to 5V rail-to-rail applications. This op amp

is engineered to amplify signals from various sensors such as temperature, pressure, and flow rate, making it suitable for use in smartphones, small IoT devices, and similar applications.



The op amp is packaged in a WLCSP with a ball pitch of just 0.3mm, reducing its size by approximately 69% compared to conventional products and 46% compared to other compact products. The TLR377GYZ enhances low input offset voltage and noise performance without compromising on miniaturisation.

ROHM

<https://www.rohm.com>

Logic-optimised general-purpose FPGAs

Lattice Semiconductor has introduced logic-optimised Lattice Certus-NX FPGA devices. The Certus-NX-28 and Certus-NX-09 models offer packaging options that improve power efficiency, size,



and reliability, with migration solutions for applications in communications, comput-

ing, industrial, and automotive sectors. These devices use less power, supporting longer battery life and simpler thermal management through FD-SOI process technology. They provide a smaller form factor, the highest I/O count per package, and cost advantages due to their low power needs and no power sequencing requirements. They also offer faster instant-on configuration performance.

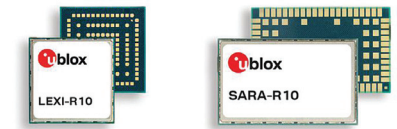
Lattice Semiconductor

<https://www.latticesemi.com>

World's smallest single-mode LTE Cat 1bis module

u-blox has announced an expansion of its R10 product line, targeting the rapidly growing LTE Cat 1bis cellular connectivity market. As 2G and 3G networks phase out worldwide, the SARA-

R10 provides a seamless upgrade path to 4G LTE, the most prevalent global cellular standard, expected to remain so for the foreseeable future. The product supports the integration of an eSIM and



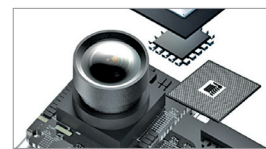
includes an integrated Wi-Fi sniffer, enabling indoor localisation through the u-blox CellLocate service, which utilises both Wi-Fi and cellular networks. This module is particularly suitable for asset tracking and telematics applications that require consistent connectivity and precise positioning both indoors and outdoors, leveraging GNSS, Wi-Fi scanning, and global LTE coverage. Moreover, this module is 50% smaller than its predecessor.

u-blox

<https://www.u-blox.com>

A machine vision sensing solution for smart applications

eYs3D Microelectronics has launched SenseLink, a multi-camera sensing system chip. SenseLink can receive signals from up to seven cameras, providing adjustable directionality and angles



for various scenarios. It includes three image signal processors, two for colour

and one for black-and-white, enabling diverse image processing within one system. It also supports metadata, enhancing data processing accuracy by transmitting data such as exposure, optical flow, and timestamps from various sensor settings. This technology is effective in smart homes, industrial automation, and medical assistance, offering robust visual sensing solutions. SenseLink facilitates fast multi-camera operations that capture images quickly, crucial for detecting changes in objects or scenes.

eYs3D Microelectronics

<https://www.eyes3d.com>